

Modeling

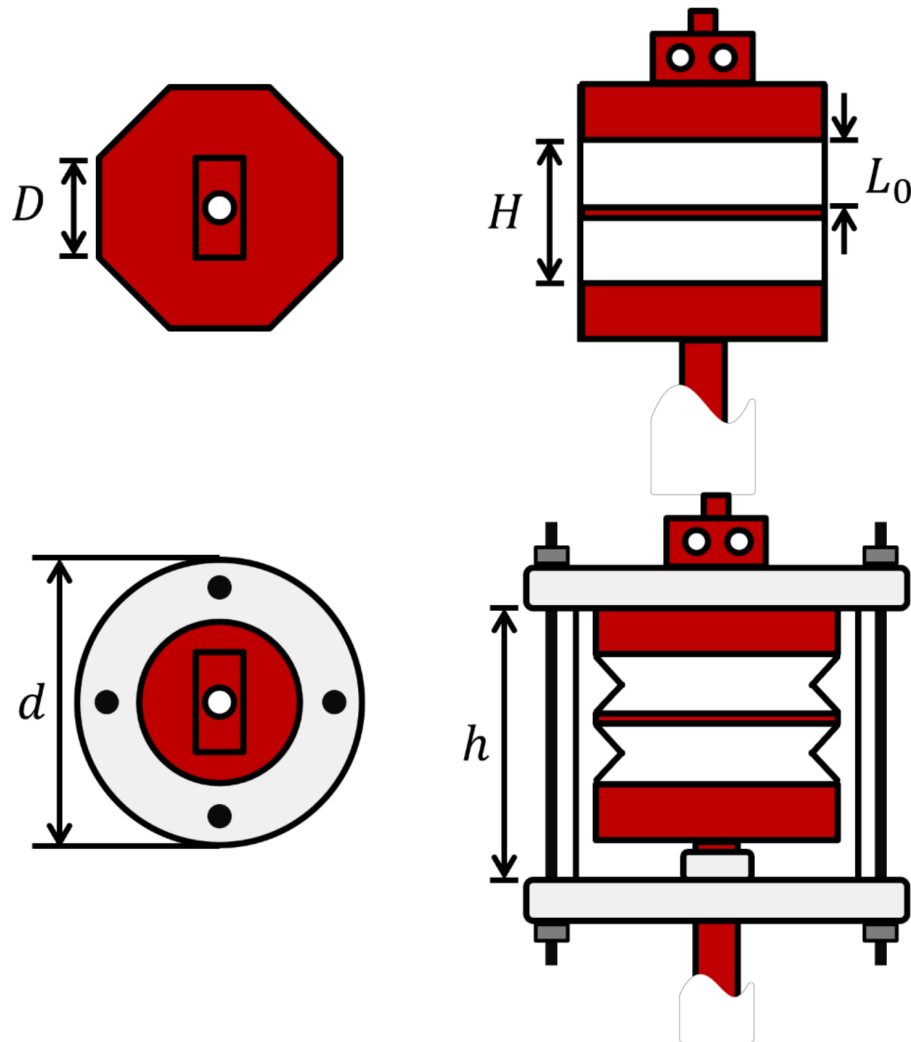


Figure 4-6. Major design parameters involved in the fabrication of the Bidirectional Hyper-VAM.

Past System

$$\dot{P}_{\text{ch}}^{\text{pos}} = \frac{RT_{\text{ch}}}{V_{\text{ch}}^{\text{pos}}} \left(\sum_{i \in \{1,2\}} |\dot{m}_{\text{pis},i}^{\text{out}}| - |\dot{m}_{\text{SV}}^{\text{pos}}| \right)$$

$$\dot{P}_{ch}^{neg} = \frac{RT_{ch}}{V_{ch}^{neg}} \left(- \sum_{i \in \{1,2\}} |\dot{m}_{pis,i}^{in}| + |\dot{m}_{SV}^{neg}| \right)$$

Current System

P_{ch} = 챔버의 압력, P_1 = Soft Actuator1 의 압력, P_2 = Soft Actuator2 의 압력

압력도 양압이나 음압이나에 따라서 분리되는 구조

V_{ch} = 챔버의 부피, V_1 = Soft Actuator1 의 부피, V_2 = Soft Actuator2 의 부피

부피도 양압이나 음압이나에 따라서 분리되는 구조

$$\dot{P}_{ch}^{pos} = \frac{RT_{ch}}{V_{ch}^{pos}} \left(\sum_{i \in \{1,2\}} |\dot{m}_{pis,i}^{out}| - |\dot{m}_{SV}^{pos}| - |\dot{m}_{SV1_1}^{pos}| - |\dot{m}_{SV2_1}^{pos}| \right)$$

$$\dot{P}_{ch}^{neg} = \frac{RT_{ch}}{V_{ch}^{neg}} \left(- \sum_{i \in \{1,2\}} |\dot{m}_{pis,i}^{in}| + |\dot{m}_{SV}^{neg}| + |\dot{m}_{SV1_1}^{neg}| + |\dot{m}_{SV2_1}^{neg}| \right)$$

$$P_1^{pos} \dot{V}_1^{pos} + \dot{P}_1^{pos} V_1^{pos} = RT_{ch} (\dot{m}_{SV1_1}^{pos} - \dot{m}_{SV1_2}^{pos})$$

$$P_2^{pos} \dot{V}_2^{pos} + \dot{P}_2^{pos} V_2^{pos} = RT_{ch} (\dot{m}_{SV2_1}^{pos} - \dot{m}_{SV2_2}^{pos})$$

$$P_1^{neg} \dot{V}_1^{neg} + \dot{P}_1^{neg} V_1^{neg} = RT_{ch} (\dot{m}_{SV1_1}^{neg} - \dot{m}_{SV1_2}^{neg})$$

$$P_2^{neg} \dot{V}_2^{neg} + \dot{P}_2^{neg} V_2^{neg} = RT_{ch} (\dot{m}_{SV2_1}^{neg} - \dot{m}_{SV2_2}^{neg})$$

$$m_{rod} \ddot{L} = -(F_{pos} + F_{neg})$$

Presumption

$$\text{Contraction Ratio} = \frac{L_0 - L}{L_0} \times 100\%$$

Contraction Ration $\leq 36.34\%$

V_f = (부채꼴 - 삼각형) * D (정8각형의 한 변의 길이)

Contraction ratio < 36.34%

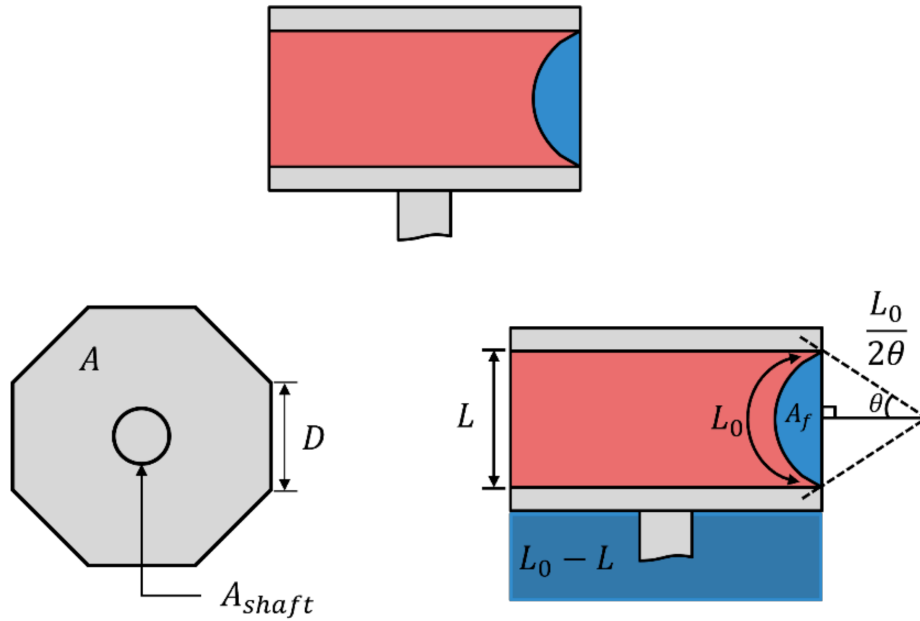


Figure 2-8. Side and top view of the change in volume before reaching a contraction ratio of 36.34%.

$$V_{oct} = A_{oct}L = 2D^2L(1 + \sqrt{2})$$

$$V_f = 8 \times A_f D = \frac{2DL_0^2}{\theta} - \frac{2DL_0^2 \sin \theta \cos \theta}{\theta^2}$$

→ 이미 8이 곱해진 형태라고 보는게 맞음. (반지름이 $\frac{L_0}{2\theta}$ 기 때문에)

$$L = \frac{L_0 \sin \theta}{\theta}$$

$$V_{neg} = n \times (V_{oct} - V_f)$$

n은 접혀있는 영역의 개수 (우리의 액추에이터에서는 2), Robotory 액추에이터에서는 더 많음

$$V_{neg}(L) = n \times \left[\frac{2D^2L_0(1 + \sqrt{2}) \sin \theta}{\theta} - \frac{2DL_0^2}{\theta} \left(1 - \frac{\cos \theta \sin \theta}{\theta} \right) \right]$$

$$\frac{dV_{neg}}{dL} = \frac{dV_{neg}}{d\theta} \times \frac{d\theta}{dL}$$

$$\frac{dV_{\text{neg}}}{dL} = n \times \left[2D^2(1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \cos 2\theta - \frac{\sin 2\theta}{\theta} \right) \right]$$

$$\dot{V}_{\text{neg}} = n \times \left[2D^2(1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \cos 2\theta - \frac{\sin 2\theta}{\theta} \right) \right] \dot{L}$$

$$F_{\text{neg}}(\theta) = -P_{\text{neg}} \frac{dV_{\text{neg}}}{dL} = -P_{\text{neg}} \times n \left[2D^2(1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \cos 2\theta - \frac{\sin 2\theta}{\theta} \right) \right]$$

$$V_{\text{pos}} = n \times [(L_0 - L)(A_{\text{oct}} - A_{\text{shaft}}) + V_f] = n \times [(L_0 - L)(A_{\text{oct}} - A_{\text{shaft}}) + 8A_f D]$$

$$V_{\text{pos}}(L) = n \times \left[(L_0 - L)(2D^2(1 + \sqrt{2}) - \pi r^2) + \frac{2DL_0^2}{\theta} \left(1 - \frac{\cos \theta \sin \theta}{\theta} \right) \right]$$

$$\frac{dV_{\text{pos}}}{dL} = \frac{dV_{\text{pos}}}{d\theta} \times \frac{d\theta}{dL}$$

$$\frac{dV_{\text{pos}}}{dL} = n \left[-[2D^2(1 + \sqrt{2}) - \pi r^2] - \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \cos 2\theta - \frac{\sin 2\theta}{\theta} \right) \right]$$

$$\dot{V}_{\text{pos}} = n \left[-(2D^2(1 + \sqrt{2}) - \pi r^2) - \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \cos 2\theta - \frac{\sin 2\theta}{\theta} \right) \right] \dot{L}$$

$$F_{\text{pos}}(\theta) = -P_{\text{pos}} \frac{dV_{\text{pos}}}{dL} = -P_{\text{pos}} \times n \left[-(2D^2(1 + \sqrt{2}) - \pi r^2) - \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \cos 2\theta - \frac{\sin 2\theta}{\theta} \right) \right]$$

$$\dot{P}_{\text{pos}} = \frac{RT \dot{m}_{\text{pos}} - P_{\text{pos}} \dot{V}_{\text{pos}}}{V_{\text{pos}}(L)}$$

$$\dot{P}_{\text{neg}} = \frac{RT \dot{m}_{\text{neg}} - P_{\text{neg}} \dot{V}_{\text{neg}}}{V_{\text{neg}}(L)}$$

$$F_{\text{tot}} = F_{\text{pos}} + F_{\text{neg}}$$

$$m_{\text{rod}} \ddot{L} = F_{\text{pos}}(L, P_{\text{pos}}) + F_{\text{neg}}(L, P_{\text{neg}}) - m_{\text{rod}} g$$

Contraction Ration > 36.34%

Contraction ratio > 36.34%

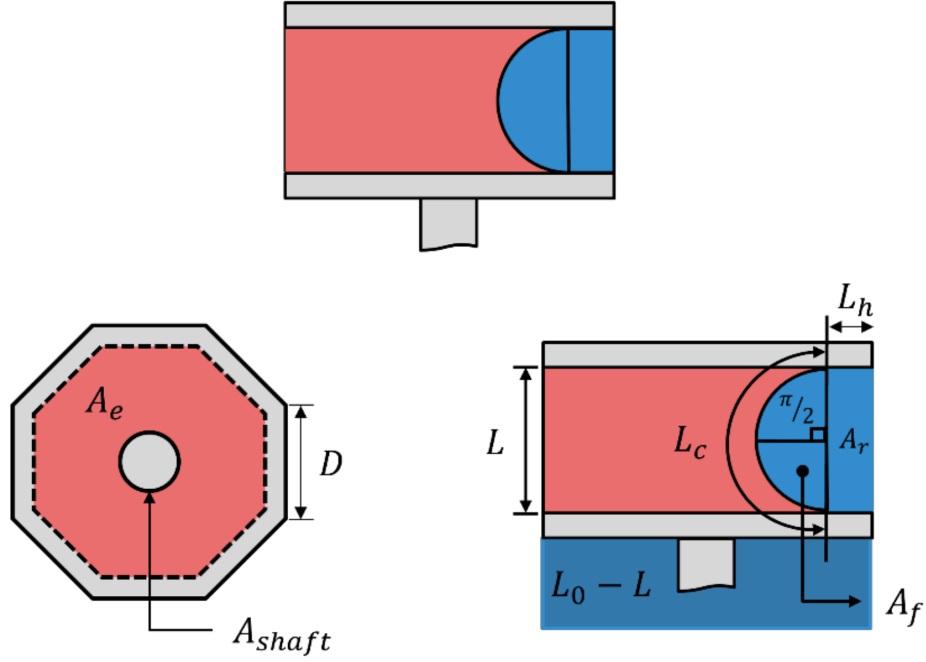


Figure 2-9. Side and top view of the change in volume after reaching a contraction ratio of 36.34%.

$$V_f = 8 \times (A_f D + A_r D) = L^2 \pi D + 8 L_h L D$$

$$L_h = \frac{2L_0 - L\pi}{4}$$

$$V_{neg}(L) = n \times (V_{oct} - V_f) = n \left[2D^2(1 + \sqrt{2})L - (L^2 D \pi + 8 L_h L D) \right]$$

$$\frac{dV_{neg}}{dL} = n \times \left[2D^2(1 + \sqrt{2}) - 4L_0 D + 2LD\pi \right]$$

$$F_{neg} = -P_{neg} \frac{dV_{neg}}{dL} = -P_{neg} \times n \left[2D^2(1 + \sqrt{2}) - 4L_0 D + 2LD\pi \right]$$

$$V_{pos} = n \times [(L_0 - L)(A_{oct} - A_{shaft}) + V_f]$$

$$V_{pos} = n \times [(L_0 - L)(2D^2(1 + \sqrt{2}) - \pi r^2) + (L^2 D \pi + 8 L_h L D)]$$

$$\frac{dV_{pos}}{dL} = n \times \left[-[2D^2(1 + \sqrt{2}) - \pi r^2] + 4L_0 D - 2LD\pi \right]$$

$$F_{\text{pos}} = -P_{\text{pos}} \frac{dV_{\text{pos}}}{dL} = P_{\text{pos}} \times n [2D^2(1 + \sqrt{2}) - \pi r^2 - 4L_0 D + 2LD\pi]$$

$$\dot{P}_{\text{pos}} = \frac{RT \dot{m}_{\text{pos}} - P_{\text{pos}} \dot{V}_{\text{pos}}}{V_{\text{pos}}(L)}$$

$$\dot{P}_{\text{neg}} = \frac{RT \dot{m}_{\text{neg}} - P_{\text{neg}} \dot{V}_{\text{neg}}}{V_{\text{neg}}(L)}$$

$$F_{\text{tot}} = F_{\text{pos}} + F_{\text{neg}}$$

$$m_{\text{rod}} \ddot{L} = F_{\text{pos}}(L, P_{\text{pos}}) + F_{\text{neg}}(L, P_{\text{neg}})$$

4th Runge Kutta

$$x = \begin{bmatrix} P_{\text{neg}} \\ P_{\text{pos}} \\ L \\ v \end{bmatrix}, \quad v = \dot{L}, \quad u = \begin{bmatrix} \dot{m}_{\text{neg}} \\ \dot{m}_{\text{pos}} \end{bmatrix}$$

$$f(x, u) = \begin{bmatrix} \frac{RT \dot{m}_{\text{neg}} - P_{\text{neg}} \dot{V}_{\text{neg}}(L, v)}{V_{\text{neg}}(L)} \\ \frac{RT \dot{m}_{\text{pos}} - P_{\text{pos}} \dot{V}_{\text{pos}}(L, v)}{V_{\text{pos}}(L)} \\ v \\ \frac{F_{\text{pos}}(P_{\text{pos}}, L) + F_{\text{neg}}(P_{\text{neg}}, L)}{m_{\text{rod}}} \end{bmatrix}$$

$$k_1 = f(x_n, u_n),$$

$$k_2 = f\left(x_n + \frac{h}{3} k_1, u_n\right),$$

$$k_3 = f\left(x_n - \frac{h}{3} k_1 + h k_2, u_n\right),$$

$$k_4 = f(x_n + h k_1 - h k_2 + h k_3, u_n),$$

$$x_{n+1} = x_n + \frac{h}{8} (k_1 + 3k_2 + 3k_3 + k_4).$$

$$t_{n+1} = t_n + h.$$